Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment MonitoringService

Baltic Sea SST

Issue: 4.0

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Approval date by the CMEMS product quality coordination team :
November 2024

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/10	All	Creation of the document	Jacob L. Høyer	
2.0	2019/07/09	All	Updated for 2018 OMI	Jacob L. Høyer	
2.1	2020/09/09	All	Updated for 2019 OMI	Jacob L. Høyer	
2.2	2021/09/10	All	Updated for 2020 OMI based on new REP product	Sotirios Skarpalezos, Wiebke Margitta Kolbe, Ioanna Karagali, Jacob L. Høyer	
3.0	2022/06/15	All	Updated for 2021 OMI	Ioanna Karagali, Wiebke Margitta Kolbe, Jacob L. Høyer	
3.1	2023/06/14	All	Updated for 2022 OMI	Ioanna Karagali, Wiebke Margitta Kolbe, Jacob L. Høyer	

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4.0	2024/05/31		Update for 2023, Change of climatology to 1991-2020, backwards extension to 1982	Ioanna Karagali, Wiebke Margitta Kolbe, Jacob L. Høyer	A.Pisano

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I INTRODUCTION

I.1 Product name

OMI_CLIMATE_sst_BAL_area_averaged_anomalies and OMI_CLIMATE_sst_BAL_trend

I.2 Indicator evaluation method

All anomalies have been calculated with respect to a monthly climatology from 1991-2020. To calculate the trends in the SST time series and the confidence intervals on the trends, we used the Mann-Kendall Tau-b non-parametric function with Sens method to compute a coefficient representing a trend for equally spaced data (Mann 1945; Sen 1968; Kendall 1975). A 95% confidence interval for the trend estimate is chosen here.

The SST Level 4 analysis product that provides the input to the monthly averages is taken from the reprocessed dataset SST_BAL_SST_L4_REP_OBSERVATIONS_010_016 with the regular updates based upon C3S SST observations. The climatology period from 1991 to 2020 (30 years) is selected according to WMO recommendations (WMO, 2017) and the most recent practice from the U.S. National Oceanic and Atmospheric Administration practice (https://wmo.int/media/news/updated-30-year-reference-period-reflects-changing-climate).

I.3 CMEMS products used as input for the computations

Product name (and Type)	Documentation
SST_BAL_SST_L4_REP_OBSERVATIONS_010_016	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-016-040-V2.pdf PUM: https://catalogue.marine.copernicus.eu/docu

	ments/PUM/CMEMS-SST-PUM-010-016-040-V2.pdf

II SCIENTIFIC VALIDATION

II.1 Global mean Ocean Heat Content

The reliability of the BALTIC SEA SST OMI products is connected to the quality and accuracy of the BAL-REP-L4 SST product (SST_BAL_SST_L4_REP_OBSERVATIONS_010_016), which is validated in the corresponding QUID.

The accuracy of the BAL-REP-L4 product (SST_BAL_SST_L4_REP_OBSERVATIONS_010_016) has been assessed by comparison to independent in situ data from drifting and moored buoys andships, for the period 1982-2021.

Results against drifting Buoys show mean bias 0.10, standard deviation 0.40, RMS 0.41 for a total of 784180 matchups.

When using moored buoys mean bias is 0.12, standard deviation 0.50, RMS 0.51for a total of 156629 matchups.

Validation statistics with ships are show a mean bias of -0.21, standard deviation0.85, RMS 0.88 for a total of 3085376 matchups.

The validation also shows a good stability of this dataset over the years,indicating that this product represents a valuable SST dataset for climate studies.

II.2 Regional maps of SST trends

SST is an essential climate variable that is an important input for initialising numerical weather prediction models and fundamental for understanding air- sea interactions and monitoring climate change. The Baltic Sea is a region that requires special attention regarding the use of satellite SST records and the assessment of climatic variability (Høyer and She 2007; Høyer and Karagali 2016). The Baltic Sea is a semi-enclosed basin with natural variability and it is influenced by large-scale atmospheric processes and by the vicinity of land. In addition, the Baltic Sea is one of the largest brackish seas in the world. When analysing regional-scale climate variability, all these effects have to be considered, which requires dedicated regional and validated SST products. Satellite observations have previously been used to analyse the climatic SST signals in the North Sea and Baltic Sea (BACC II Author Team 2015; Lehmann et al. 2011). Recently, Høyer and Karagali (2016) demonstrated that the Baltic Sea had warmed 1-2 °C from

1982 to 2012 considering all months of the year and 3-5 °C when only July-September months were considered. This was corroborated in the Ocean State Reports (section 1.1 in Von Schuckmann et al., 2016; section 3 in Von Schuckmann et al., 2018).

SST trends were calculated for the Baltic Sea area and the whole region including the North Sea, over the period January 1982 to December 2023 (Fig 1).The average trend for the Baltic Sea domain (east of 9°E longitude) is 0.038 °C/year, which represents an average warming of 1.60 °C for the 1982-2023 period considered here. When the North Sea domain is included, the trend decreases to 0.029°C/year corresponding to an average warming of 1.22°C for the 1982-2023 period. Trends are highest for the Baltic Sea region, the North Sea andNorthAtlantic, especially offshore Norway, compared to other regions.

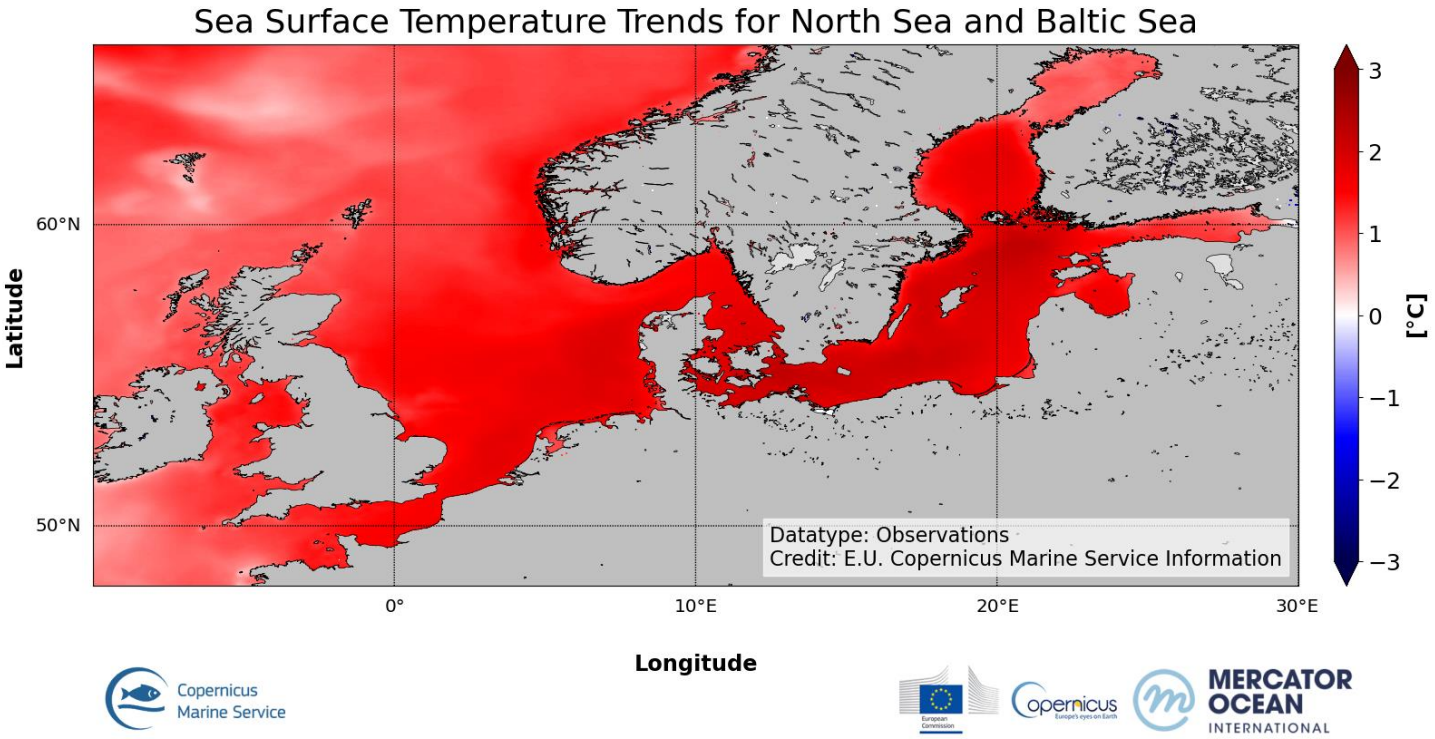

Figure 1: Cumulative trends in sea surface temperature anomalies calculated from 1982 to 2023for the Baltic Sea (OMI_CLIMATE_sst_BAL_trend). Trend calculations are based on the multi-year Baltic Sea L4 SST satellite product SST_BAL_SST_L4_REP_OBSERVATIONS_010_016.

III REFERENCES

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